



University of
Nottingham

UK | CHINA | MALAYSIA

COMP2001/2011: Artificial Intelligence Methods

Attendance Monitoring
QR Code

Lecture 11 – Revision

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Announcements for COMP2001 Students

- In-lab practical exam this **Friday 8th May**.
- Check your room and time slot on Moodle.
- Some points to remember for the in-lab practical exam:
 - You are not allowed to use any AI tools – disable them if active in your IDE.
 - You must use your implementation submitted to Moodle along with the same configuration/parameter settings – we will re-run a sample of submissions.
 - Use the version of HyFlex and set the duration appropriate to the machine you are taking the exam on (original vs wall timer JARs).



Format of Exam

- 1-hour ExamSys exam.
- All questions are (dropdown; multiple choice; numerical input; true/false) in a style similar to the weekly formative quizzes.
- No negative marking – if you are unsure, make a best guess!

School of Computer Science

A Level 2 Module Spring 2025/26

Artificial Intelligence Methods

Instructions:

Time allowed: ONE Hour

Answer ALL questions

Candidates may now sign their desk card.

Only a calculator from approved list A may be used in this examination.

Basic Models

Scientific Calculators

	Aurora AX-582	Casio FX82 family
Aurora HC133	Casio FX83 family	Casio FX85 family
Casio HS-5D	Casio FX350 family	Casio FX570 family
Deli – DL1654	Casio FX 991 family	Sharp EL-531 family
Sharp EL-233	Texas Instruments TI-30 family	Texas BA II+ family

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Appended material: NONE

Additional material: NONE

Information for Invigilators: NONE

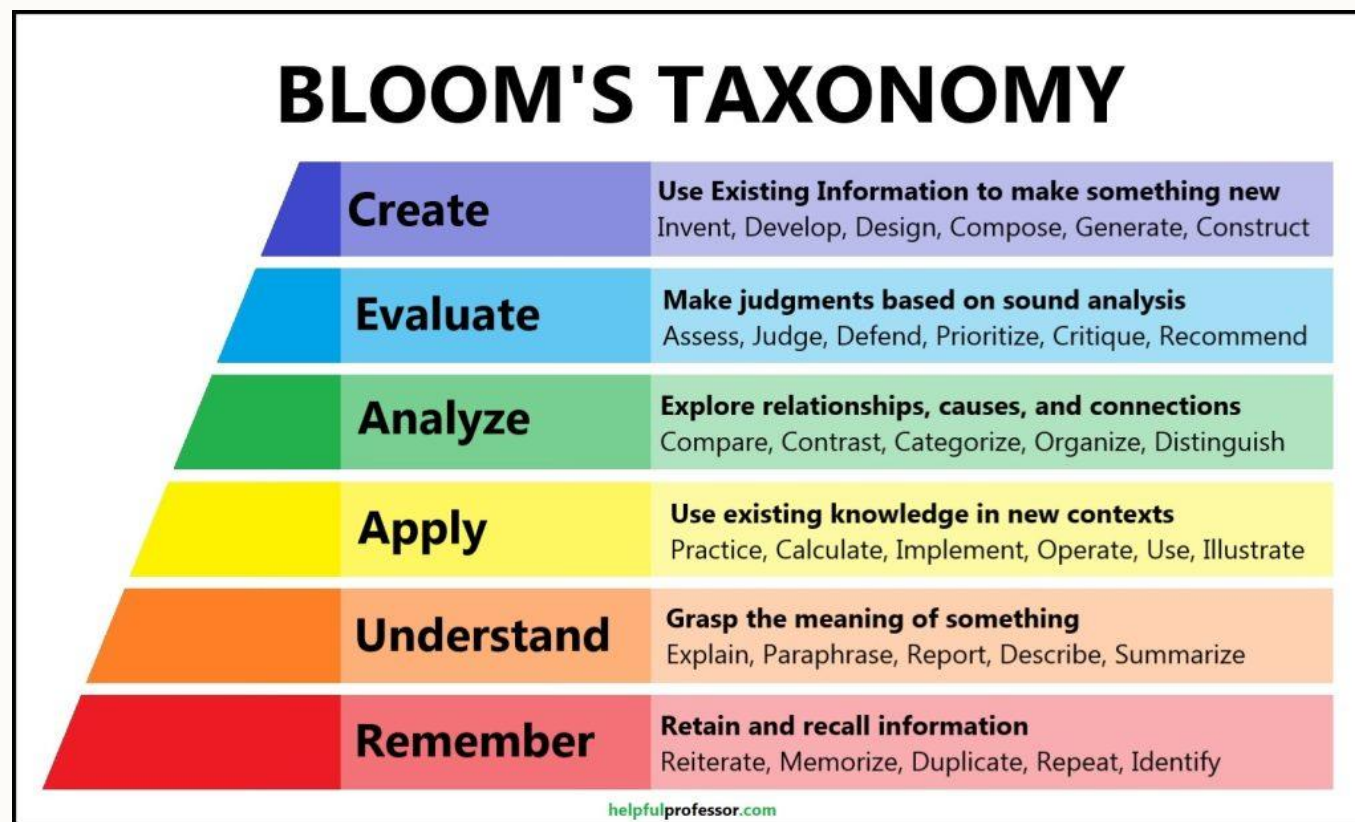
Exam Papers must not be removed from the examination venue.

You are not permitted to take any paper or notes out of the examination room during or after the examination. Any rough notes that you make on the paper provided will be collected by staff and destroyed.



Examinable Material

- Anything that was written/spoken/discussed in lectures/Moodle materials/Moodle forums.
- Types of questions:
 - **Knowledge** – questions to assess factual understanding of the content.
 - **Comprehension** – questions to assess understanding of concepts.
 - **Application** – questions that require you to apply your knowledge from the course to solve new tasks.





Recap and Interactive Quiz

Lecture 1 – Introduction and search paradigms



What we learned – Lecture 1

- Introduction to search / optimisation / decision support.
- Search paradigms
 - Exact vs in-exact
 - Discrete vs continuous
 - Single-objective vs multi-objective
 - ...
- Heuristic methods with an application to **Bin Packing** (e.g. LIFF).
- A case study on perturbative and constructive heuristics for **Travelling Salesman Problem (TSP)**.
- Pseudo-random number generators.



Optimisation Problems usually consist of which of the following?

- A. Constraints
- B. Initialisation method
- C. Metaheuristics
- D. Objective function



Exact methods are preferred over in-exact methods because...

- A. In-exact methods cannot find the optimal solution.
- B. In-exact methods do not return good quality solutions.
- C. In-exact methods have no optimality guarantees.
- D. In-exact methods may never terminate.



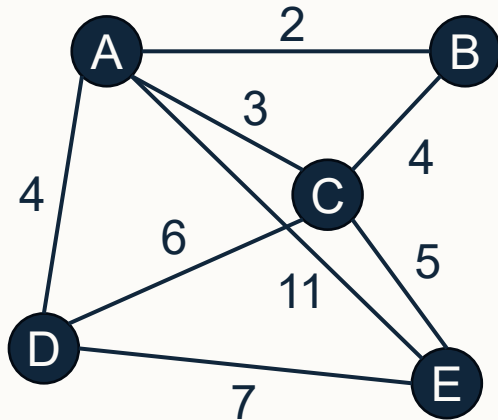
Which heuristic(s) are guaranteed to optimally pack the following items with sizes $\{5, 3, 1, 2, 4\}$ in a Bin Packing problem with $V = 5$?

- A. Largest Item First Fit
- B. Smallest Item First Fit
- C. Random First Fit
- D. Best fit (sequential ordering)



What solution would be obtained to the following TSP instance after applying nearest neighbour with city A being selected as the starting city? Enter your solution in the form ABCDE

Rank Response





Recap and Interactive Quiz

Lecture 2 – Heuristic search and hill-climbing



What we learned - lecture 2

- More combinatorial optimisation problems:
 - 0/1 Knapsack Problem
 - Boolean Satisfiability Problem
- Components of heuristic search methodologies.
 - Guidelines
 - Representations
 - Neighbourhood relations / move operators
 - Objective functions / delta evaluation
 - (Initialisation methods / mechanisms for escaping local optima)
- Hill climbing methods
 - Random / First Improvement / Best Improvement / Davis's Bit



Which of the following representations is most suitable for encoding solutions to the Knapsack Problem?

- A. Binary Encoding
- B. Integer Encoding
- C. Permutation Encoding
- D. Value Encoding
- E. Other



Select all the move operators that are guaranteed to produce a candidate solution which has a Hamming Distance of 1 for a KP.

- A. Random Bit Flip
- B. Adjacent Swap
- C. (Steepest/Best) descent
- D. Next descent/first improvement
- E. Davis's Bit Hill Climbing
- F. 1-point crossover



The multiple knapsack problem (MKP) extends the standard knapsack problem by selecting a subset of n items and packing them into a fixed number m of capacity-constrained knapsacks to maximise the total value. Which representation would you choose to encode solutions to the MKP?

- A. Binary encoding
- B. Integer encoding
- C. Permutation encoding
- D. Value encoding
- E. Other



You are designing a neighbourhood relation for solving the MKP using integer encoding as the representation. Select one or more of the following that have the largest neighbourhood size?

- A. Remove random item.
- B. Swap random items.
- C. Increment random item.
- D. Pack new item into least filled KP.



Recap and Interactive Quiz

Lectures 3 & 4 – Metaheuristics and Move Acceptance



What we learned - lectures 3 & 4 (1)

- Metaheuristics (single-point perturbative/local search)
 - Mechanisms for escaping local optima
 - Termination criteria
 - Dealing with infeasibility
- Examples of metaheuristics for solving MAX-SAT / TSP
 - Iterated Local Search
 - Tabu Search
- Performance comparisons of algorithms
 - Statistical tests / ranking methods / boxplots
- Scheduling: concepts and notation



What we learned - lectures 3 & 4 (2)

- Move acceptance
 - Methods and taxonomy
 - Late Acceptance / Simulated Annealing
- Parameter setting issues
 - Influence of parameter settings on search performance
- Parameter tuning / control
- Methods to perform parameter tuning
 - Sequential tuning
 - Fractional arrays
 - Taguchi OAs



What is the total weighted tardiness for a scheduling problem $1 \mid p_j, d_j, w_j, r_j \mid \sum w_j T_j$ where there are three jobs to be processed on a single machine with the schedule $\langle 1, 3, 2 \rangle$ where the jobs $j = 1, \dots, 3$ have the characteristics as shown on the slides:

Jobs	1	2	3
p_j	2	3	4
d_j	3	4	10
w_j	2	2	1
r_j	0	5	5



Eight greedy heuristics are evaluated for solving multiple instances of the TSP. Select the heuristic(s) that perform the best for solving small TSP instances.

A. H1

B. H2

C. H3

D. H4

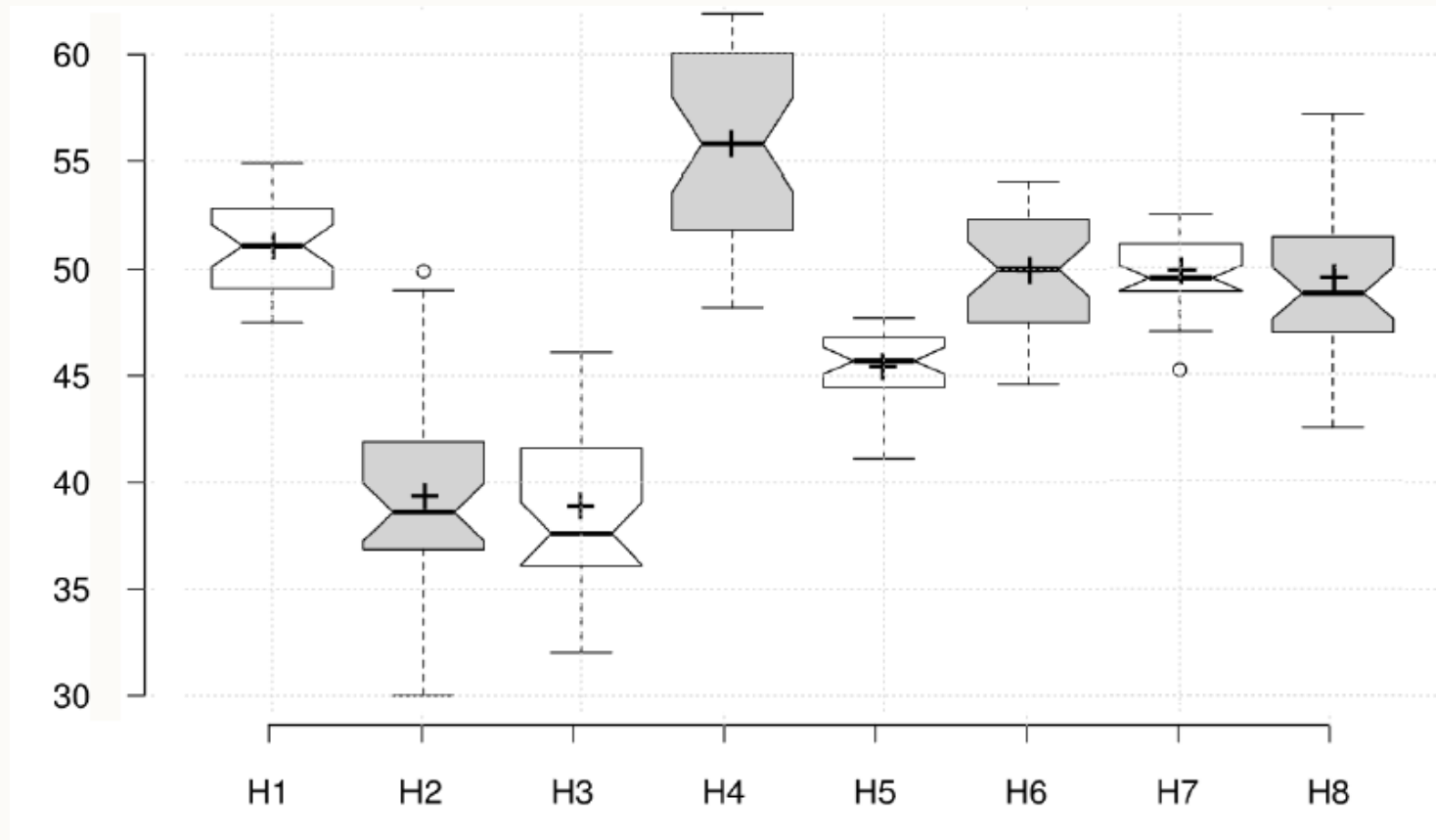
E. H5

F. H6

G. H7

H. H8

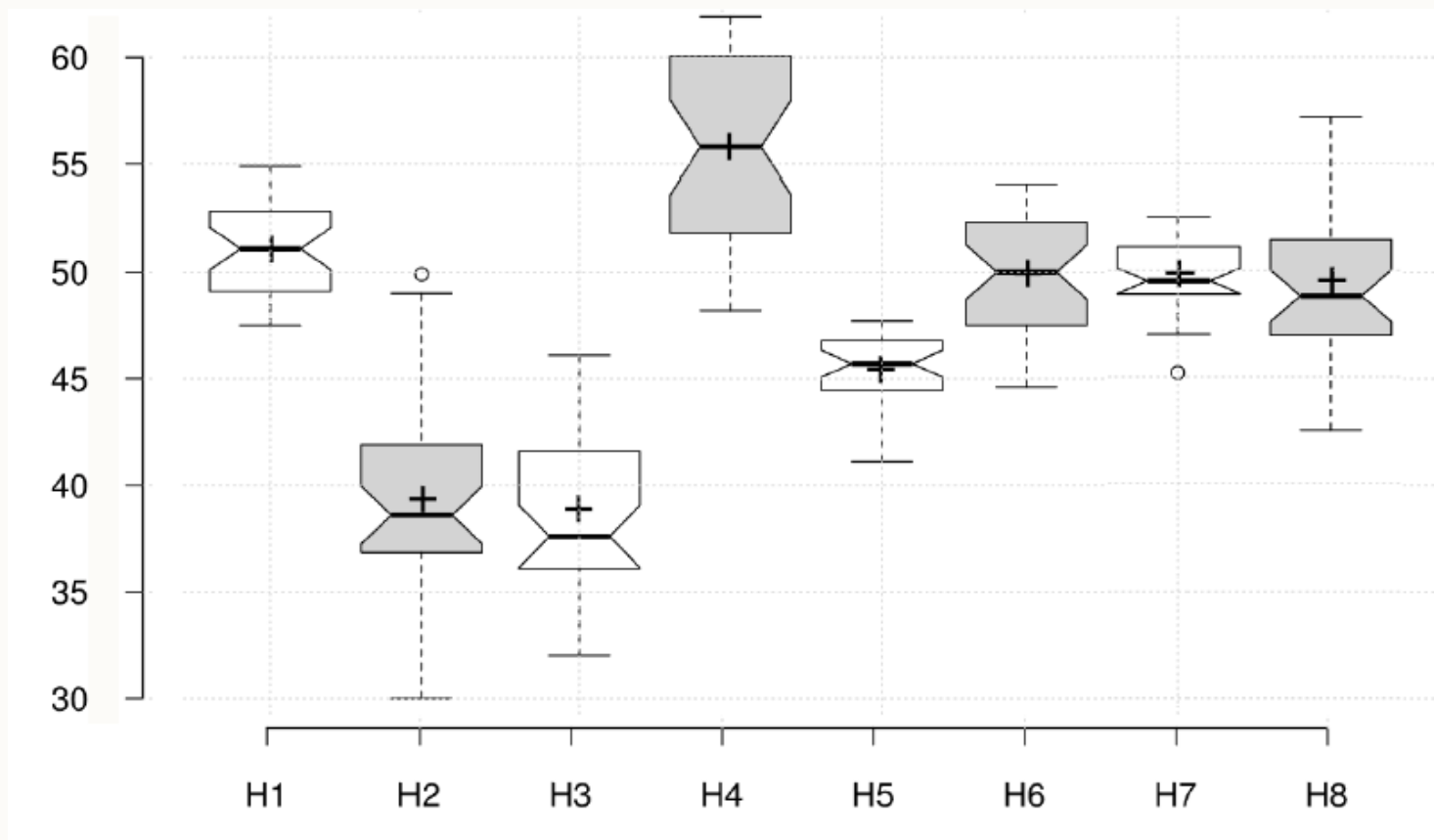
I. Not possible to say





Eight greedy heuristics are evaluated for solving an instance of the TSP. Select all heuristic(s) that perform statistically significantly differently than H_8 .

1. H1
2. H2
3. H3
4. H4
5. H5
6. H6
7. H7





You develop a new metaheuristic for solving vehicle routing problems. You want to know whether your algorithm is the new state of the art by comparing against results obtained from an online archive of benchmark results from the current state-of-the-art method. Given the limited information above, which of the following statistical tests should definitely not be used to test your hypothesis?

- A. Comparison of mean averages
- B. Friedman Test
- C. Mann-Whitney U Test
- D. Wilcoxon Signed Rank Test



Select all the move acceptance methods that are able to accept worse cost moves.

- A. Iterated Local Search
- B. Late Acceptance
- C. Naïve Acceptance
- D. Simulated Annealing



Select all the move acceptance methods that could be configured to never accept worse cost moves.

- A. Late Acceptance
- B. Naïve Acceptance
- C. Record-to-Record Travel
- D. Simulated Annealing



Recap and Interactive Quiz

Lectures 5 & 6 – Evolutionary Algorithms

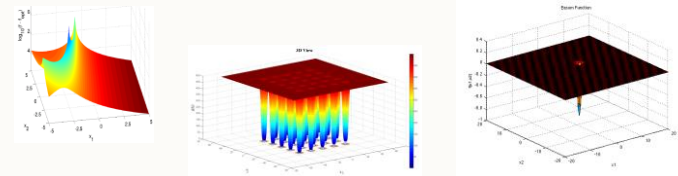


What we learned – lecture 5

- Evolutionary Analogy
- Genetic Algorithm (GA) Components
 - Representation
 - Initialisation
 - Fitness Evaluation
 - Selection (i) Roulette Wheel (ii) Tournament Selection
 - Crossover (Recombination) (i) 1-Point Crossover (ii) Uniform Crossover
 - Mutation (i) bit-flip
 - Replacement Strategy (i) Trans-Generational (ii) Steady-State (-) Elitism
 - Termination Criteria
- Memetic Algorithms (MA)
 - Hill climbing



What we learned – lecture 6



- Benchmark Functions
 - Continuity (Differentiability)
 - Scalability
 - Separability
 - Modality (i) Unimodal (i) Multimodal
- Case Studies
 - GA vs. MA
- Permutation-based Genetic Operators
 - PMX, OX, CX
- Multimeme Memetic Algorithms (MMA)
 - Memplex
- Case Studies
 - GA vs MA vs MMA



Which of the following generic operators in a memetic algorithm are most likely to worsen the fitness of the output solution(s) than the input solution(s) when applied? (Select all that apply.)

- A. Crossover
- B. Mutation
- C. Hill Climbing
- D. Replacement with Elitism

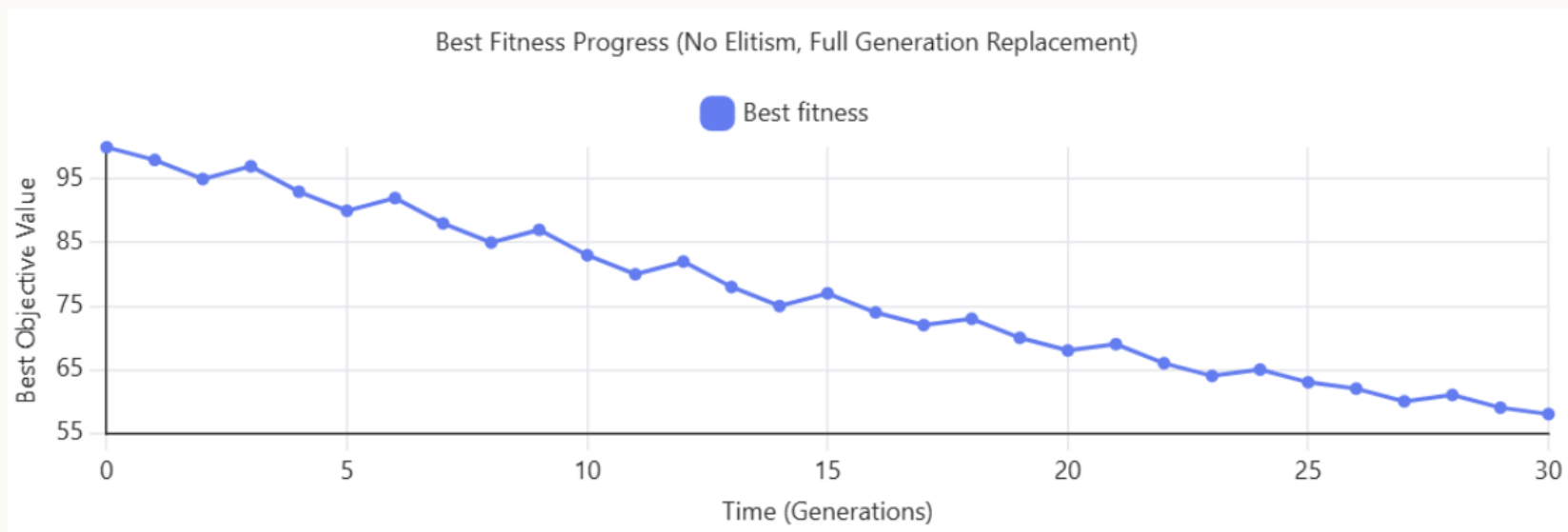


Which of the following statements about Genetic Algorithms (GAs), Memetic Algorithms (MAs), and Multimeme Memetic Algorithms (MMAs) are TRUE? (Select all that apply.)

- A. GAs perform global exploration via its operators.
- B. MMAs attempt to combine multiple GAs, each using a different mutation rate, to achieve adaptation.
- C. MMAs extend MAs by enabling adaptive selection or evolution of local search strategies (memes) during a run.
- D. MMAs differ primarily by using specialised representations, such as permutation, that GAs and MAs cannot handle effectively.



You are given a progress plot showing the best individual's fitness over time from a single run of an evolutionary algorithm on a problem instance. Which of the following statements could be TRUE? (Select all that apply.)



- A. This algorithm cannot be a memetic algorithm.
- B. This algorithm could be a genetic algorithm.
- C. The replacement scheme is based on full generational replacement (generation gap = population size).
- D. The replacement scheme is not elitist at all.



Which of the following candidate solutions are valid offspring after applying one-point crossover to the parents 111111 and 000000?

- A. 111000
- B. 001111
- C. 111100
- D. 111101
- E. 101111



Assume standard definitions of the crossover operators as discussed in class. Which of the following crossover operators could produce the offspring “111101” and “000010” when applied to parents “111111” and “000000”? (Select all that apply.)

- A. 1-point crossover
- B. 2-point crossover
- C. K-point crossover (where k is decided randomly)
- D. Uniform crossover



What does an innovation rate of 0.2 mean when used in an Multimeme Memetic Algorithm?

- A. Exactly 20% of memes are removed each generation.
- B. A solution adopts a new meme with probability 0.2.
- C. Only 20% of the population uses memes.
- D. Meme adaptation stops after 20 generations.



Recap and Interactive Quiz

Lectures 7 & 8 – Hyper-heuristics



What we learned - lectures 7 & 8 (1)

- Hyper-heuristics (selection and generative)
- HyFlex and Cross-domain search
- Methods for heuristic selection
 - Non-learning methods:
 - Random
 - Greedy
 - Learning-based methods:
 - Reinforcement learning
 - Choice function
- Parameter tuning of cross-domain search methods

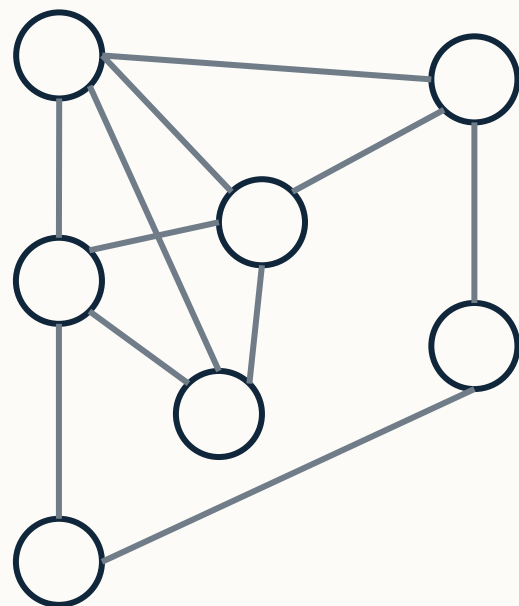


What we learned - lectures 7 & 8 (2)

- Graph-based hyper-heuristics
- Graph colouring (including graph heuristics)
- Genetic programming
- Examination timetabling



An instance of an exam timetabling problem can be represented using the following graph. What is the minimum number of time periods required to schedule all exams such that no student has more than one exam at any time?

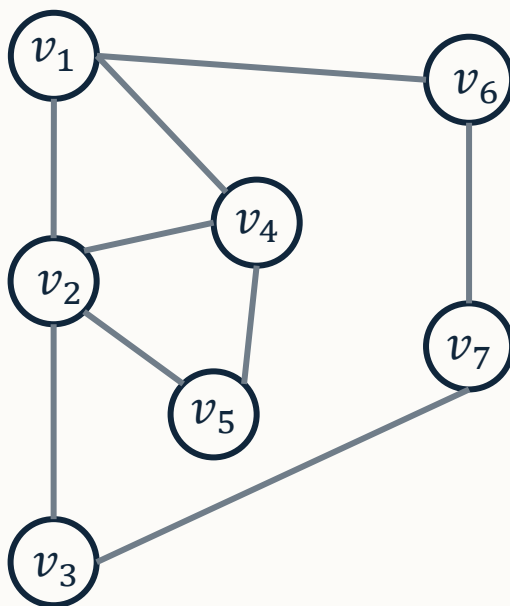


Reminder: nodes are exams, and edges represent exams that have common students.



Apply the largest degree heuristic to the following graph assuming that you can choose from the colours $\{r, g, b, y, c, m\}$. For tie breaking, choose the vertex v_n over v_m whereby $n < m$. Which colour is used to colour vertex V_5 ?

- A. Red
- B. Green
- C. Blue
- D. Yellow
- E. Cyan
- F. Magenta





Scenario: 3D Production Planning Problem (3DPPP)

In a 3DPPP, there are n parts to be 3D printed into i build volumes over j consecutive days. For simplicity we assume a build volume can be 3D printed in no more than 24 hours and there is no changeover time. This problem can be described as: $Pi \mid r_n, d_n \mid \sum T_n$.

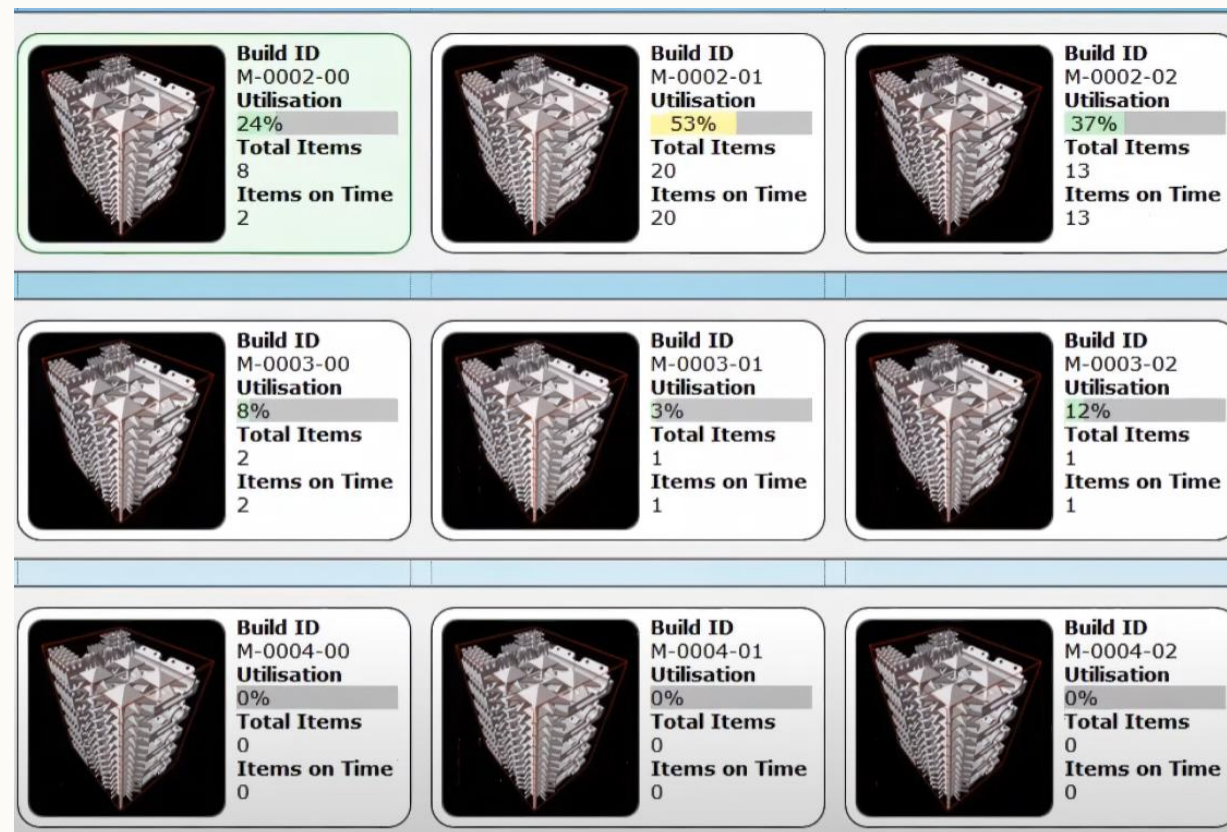
Each part can be packed into any build volume $V_{i,j}$ at a given 3D coordinate (x, y, z) describing its Bottom-Left-Front most position as a bounding box.

Constraints:

- Parts cannot be printed before r_n .
- Parts cannot intersect.
- Parts must fit wholly within the build volume.

Note that:

- Parts may “float in space” and do not require support structures.





You are designing a constructive selection hyper-heuristic for solving the 3DPPP. You design three constructive low-level heuristics, each of which packs a single un-packed item into a single build volume $V_{i,j}$ (their exact details are unimportant). Which of the representations is most suitable?

- A. Binary Encoding
- B. Integer Encoding
- C. Permutation Encoding
- D. None of the above as neither is a complete representation for this problem.



You are designing a constructive selection hyper-heuristic for solving the 3DPPP. You design three constructive low-level heuristics, each of which packs a single un-packed item into a single build volume $V_{i,j}$ (their exact details are not important). Given that a solution can be encoded using integer encoding, what would be the length of the array encoding a solution?

- A. n*
- B. ij*
- C. nij*
- D. $3nij$*



You are designing a constructive selection hyper-heuristic for solving the 3DPPP. You design three constructive low-level heuristics, each of which packs a single un-packed item into a single build volume $V_{i,j}$ (their exact details are unimportant). Given that a solution can be encoded using integer encoding, name a suitable method that could be used for solution initialisation.



You are designing a constructive selection hyper-heuristic for solving the 3DPPP. You design three constructive low-level heuristics, each of which packs a single un-packed item into a single build volume $V_{i,j}$ (their exact details are unimportant). Given that a solution can be encoded using integer encoding, name a suitable neighbourhood operator that could be used to search the space of constructive heuristics for this approach.



Thank you

Remainder of lecture will be Q&A – any questions?